



DYNAMIC STOPPING DISTANCE PHYSICS (d_{stop})

Translating Milliseconds of Information Staleness into Physical Centimeters of Collision Hazard

$$d_{\text{stop}}(t) = \underbrace{v(t) \cdot t_{\text{delay}}(t)}_{\text{1. Reaction Distance (Linear in Latency } \Delta t\text{)}} + \underbrace{\frac{v(t)^2}{2 \cdot a_{\text{max}}}}_{\text{2. Braking Distance (Quadratic in Velocity } v\text{)}}$$

✓ **Nominal Case** ($v = 1.0 \text{ m/s}$, $t_{\text{delay}} = 30 \text{ ms}$, $a_{\text{max}} = 2.0 \text{ m/s}^2$):

$d_{\text{react}} = 0.03 \text{ m}$

$d_{\text{brake}} = 0.25 \text{ m}$

Total $d_{\text{stop}} = 0.28 \text{ m}$ (Safe: Clearance = 0.35 m)

⊘ Physical Obstacle Clearance Barrier (0.35 m)

⚠ **Tail Spike Case** ($v = 1.0 \text{ m/s}$, $t_{\text{delay}} = 230 \text{ ms}$, $a_{\text{max}} = 2.0 \text{ m/s}^2$):

$d_{\text{react}} = 0.23 \text{ m}$ (+20 cm Blind Travel)

$d_{\text{brake}} = 0.25 \text{ m}$

$d_{\text{stop}} = 0.48 \text{ m}$ (CRASH!)